

YouTube Content Monetization Modeler

Interpretation, Insights & Documentation

1. Project Overview

This project predicts YouTube advertisement revenue (`ad_revenue_usd`) using machine learning models based on video engagement, audience behavior, and content-related features.

The workflow includes:

1. Data loading and preprocessing
 2. Feature engineering
 3. Encoding categorical variables
 4. Train-test split
 5. Feature scaling
 6. Training multiple regression models
 7. Model evaluation
 8. Saving the best model
 9. Building a Streamlit dashboard for predictions
-

2. Dataset Features

The dataset contains the following important features:

Feature	Description
views	Total video views
likes	Total likes received
comments	Total comments received
watch_time	Total audience watch time
video_length	Length of the video
subscribers	Channel subscriber count
category	Video category
device	Device used by viewers

Feature	Description
country	Audience country
engagement_rate	Engineered feature calculated using likes and comments

Target Variable:

- `ad_revenue_usd` → Estimated advertisement revenue

3. Data Preprocessing

Duplicate Removal

```
df.drop_duplicates(inplace=True)
```

Purpose:

- Removes repeated rows from the dataset.
- Prevents bias during model training.

Handling Missing Values

```
df.fillna(method='ffill', inplace=True)
```

Purpose:

- Fills missing values using forward fill.
- Ensures the dataset contains no null values before training.

4. Feature Engineering

A new feature called `engagement_rate` was created.

```
df['engagement_rate'] = (df['likes'] + df['comments']) / df['views']
```

Why Engagement Rate Matters

Engagement rate measures how actively users interact with the content.

Higher engagement generally indicates:

- Better audience retention
- Higher ad interaction potential
- Improved monetization opportunities

This engineered feature helps the model better understand audience quality rather than only audience quantity.

5. Encoding Categorical Features

```
le = LabelEncoder()

df['category'] = le.fit_transform(df['category'])
df['device'] = le.fit_transform(df['device'])
df['country'] = le.fit_transform(df['country'])
```

Purpose

Machine learning algorithms require numerical data.

Label Encoding converts text categories into numeric representations.

Examples:

Original Category	Encoded Value
Music	0
Gaming	1
Education	2

6. Feature Selection

```
X = df.drop(['ad_revenue_usd', 'video_id', 'date'], axis=1)
y = df['ad_revenue_usd']
```

Explanation

Input Features (X)

Used for prediction.

Target Variable (y)

Revenue value to predict.

Dropped Columns

Column	Reason
ad_revenue_usd	Target variable
video_id	Unique identifier with no predictive value
date	Not directly useful in current model

7. Train-Test Split

```
train_test_split(X, y, test_size=0.2)
```

Purpose

- 80% data used for training
- 20% data used for testing

This helps evaluate how well the model performs on unseen data.

8. Feature Scaling

```
scaler = StandardScaler()
```

Why Scaling is Important

Features such as:

- views
- subscribers
- watch_time

may have very large numerical ranges.

Scaling standardizes the data and improves model performance, especially for:

- Linear Regression
 - SVR
-

9. Models Used

The following regression models were trained:

Model	Purpose
Linear Regression	Baseline regression model
Random Forest Regressor	Ensemble learning model
Decision Tree Regressor	Tree-based regression
SVR	Support Vector Regression

10. Model Evaluation Metrics

The models were evaluated using:

R² Score

Measures how well the model explains variance.

- Closer to 1 → Better performance
-

RMSE (Root Mean Squared Error)

Measures prediction error magnitude.

- Lower RMSE → Better accuracy
-

MAE (Mean Absolute Error)

Average absolute prediction error.

- Lower MAE → Better model reliability
-

11. Best Performing Model

The Random Forest Regressor was selected as the best model.

Why Random Forest Performed Better

Random Forest:

- Handles non-linear relationships effectively
 - Reduces overfitting using multiple trees
 - Captures complex interactions between features
 - Performs well on mixed numerical and categorical datasets
-

12. Interpretation & Insights

Most Influential Features on Revenue

Based on the dataset structure and Random Forest behavior, the following features are likely to have the strongest influence on advertisement revenue:

1. Views

- Revenue increases as video views increase.
- More views lead to more ad impressions.
- This is typically the strongest revenue-driving feature.

2. Watch Time

- Longer watch time improves monetization potential.
- Videos with higher retention can display more advertisements.
- Indicates strong audience engagement.

3. Subscribers

- Channels with more subscribers usually generate consistent traffic.
- Loyal audiences improve repeat viewership and ad performance.

4. Engagement Rate

- High engagement signals quality content.
- Videos with more likes/comments relative to views often perform better in recommendation systems.
- Increased visibility indirectly improves revenue.

5. Likes and Comments

- Positive interaction metrics indicate audience satisfaction.
- These features contribute to platform promotion and organic reach.

6. Category

Different content categories generate different CPM (Cost Per Mille) rates.

For example:

- Finance and Technology often have higher ad revenue.
- Entertainment content may receive high views but lower CPM.

7. Country

Revenue varies significantly across countries.

Audiences from countries with higher advertiser spending typically produce:

- Higher CPM
- Better ad revenue

8. Device

Viewer devices can influence ad delivery effectiveness.

For example:

- Desktop viewers may generate higher ad interaction rates.
 - Mobile traffic may produce different monetization behavior.
-

13. Business Insights

Key Findings

Insight 1: Engagement Matters More Than Raw Popularity

Videos with strong audience interaction often outperform videos with only high view counts.

Insight 2: Retention Directly Impacts Revenue

Longer watch time allows more advertisements to be shown.

Creators should focus on:

- Storytelling
 - Audience retention
 - Content quality
-

Insight 3: Audience Geography Affects Earnings

Creators targeting high-CPM countries can significantly improve monetization.

Insight 4: Consistent Subscriber Growth Helps Stability

Channels with larger subscriber bases show more predictable revenue patterns.

Insight 5: Content Category Influences Profitability

Some niches are more advertiser-friendly than others.

Choosing profitable categories can improve overall revenue.

14. Streamlit Dashboard

The Streamlit application allows users to:

- Enter video statistics
- Predict estimated ad revenue
- Interact with the trained model through a web interface

Main Features:

- User-friendly interface
 - Real-time prediction
 - Sidebar navigation
 - Input validation
-

15. Model Saving

```
pickle.dump(best_model, open("model/revenue_model.pkl", "wb"))  
pickle.dump(scaler, open("model/scaler.pkl", "wb"))
```

Purpose

The trained model and scaler are saved so they can be reused without retraining.

Benefits:

- Faster deployment
 - Easy integration with Streamlit
 - Portable ML workflow
-

16. Conclusion

This project successfully demonstrates how machine learning can predict YouTube advertisement revenue using engagement and audience-related metrics.

The Random Forest Regressor achieved the best performance due to its ability to model complex relationships between features.

The project also highlights the importance of:

- Audience engagement
- Watch time
- Subscriber growth
- Content strategy

The Streamlit dashboard makes the solution interactive and deployment-ready.

17. Future Improvements

Possible future enhancements:

1. Hyperparameter tuning
 2. XGBoost or LightGBM implementation
 3. Deep learning regression models
 4. Real-time YouTube API integration
 5. Advanced feature engineering
 6. Revenue trend forecasting
 7. Explainable AI using SHAP values
 8. Interactive analytics dashboard
-

18. Technologies Used

Technology	Purpose
Python	Programming language
Pandas	Data processing
NumPy	Numerical operations
Scikit-learn	Machine learning
Matplotlib / Seaborn	Data visualization
Streamlit	Web dashboard
Pickle	Model serialization

19. Final Outcome

The project provides:

- Accurate revenue prediction
- Business-oriented insights
- Interactive deployment
- End-to-end machine learning workflow
- Practical monetization analysis for creators